# FRED TALKS

14th July 2026

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# Porting 100k lines from TypeScript to Rust using Claude Code in a month

VJEUX · 26 JAN 2026 · SOURCE

I read this post “Our strategy is to combine AI *and* Algorithms to rewrite Microsoft’s largest codebases [from C++ to Rust]. Our North Star is ‘1 engineer, 1 month, 1 million lines of code.” and it got me curious, how difficult is it really?

I've long wanted to build a competitive Pokemon battle AI after watching a lot of WolfeyVGC and following the PokéAgent challenge at NeurIPS. Thankfully there's an open source project called "Pokemon Showdown" that implements all the rules but it's written in JavaScript which is quite slow to run in a training loop. So my holiday project came to life: let's convert it to Rust using Claude!

## Escaping the sandbox

Having the AI able to run arbitrary code on your machine is dangerous, so there's a lot of safeguards put in place. But... at the same time, this is what I want to do in this case. So let me walk through the ways I escaped the various sandboxes.

### git push

Claude runs in a sandbox that limits some operations like ssh access. You need ssh access in order to publish to GitHub. This is very important as I want to be able to check how the AI is doing from my phone while I do some other activities 😉

```
● Bash(git pull --rebase && git apply /tmp/prng-fix.patch)
  └ Error: Exit code 1
    ssh: connect to host github.com port 22: Operation not permitted
    fatal: Could not read from remote repository.

    Please make sure you have the correct access rights
    and the repository exists.
```

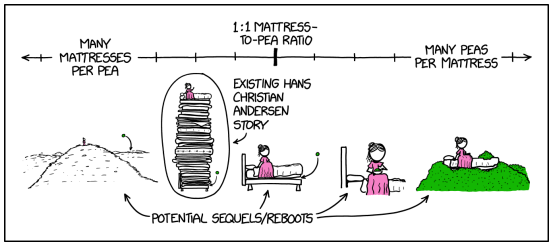
What I realized is that I can run the code on my terminal but Claude cannot do it from its own terminal. So what I did was to ask Claude to write a nodejs script that opens an http server on a local port that executes the git commands from the url. Now I just need to keep a tab open on my terminal with this server active and ask Claude to write instructions in Claude.md for it to interact with it.


### Acknowledgements

Thanks to Den Delimarsky, David Soria Parra, Henry Shi, Felix Rieseberg, Conor Kelly, Molly Vorwerck, Andy Schumeister, Kevin Garcia, Amie Rotherham, Matt Samuels, Angela Jiang, Katelyn Lesse, AJ Rebeiro and Jess Yan for their contributions to this blog.


# The Princess and the Pea

XKCD.COM · 14 JUL 2026 · SOURCE




Once we've fully explored this space, we can start varying the number of princesses.

Combining MCP servers with skills allows Claude to act more like a domain-specialist. Grab your tools via MCP, and give Claude the skills to orchestrate workflows end-to-end. See our data plugin for Cowork as an example, which consists of 10 skills and 8 MCP servers for apps like Snowflake, Databricks, BigQuery, Hex and more.

Combining skills with MCP. Source: Extending Claude's capabilities with skills and MCP servers

### Distribute skills from an MCP server

It's increasingly common for providers to publish a skill alongside their MCP server, so the agent gets both the raw capabilities and an opinionated playbook for using them well. Canva, Notion, Sentry, and more do this today in Claude, listing the skill next to their connector in our web directory.

To make that pairing portable across every client, the MCP community is actively working on an extension for delivering skills directly from servers. This way the client inherits the relevant expertise automatically, versioned with the API it depends on. We expect this pattern to see broad adoption as the extension stabilizes.

## The compounding layer

We opened with three paths for connecting agents to external systems. In practice, mature integrations will ship all three: the API as the foundation, a CLI for local-first environments, and MCP for cloud-based agents.

As production agents move to the cloud, MCP becomes the critical layer, and it's the one that compounds. Today, a remote server reaches every compatible client across any deployment environment, with auth, interactivity, and rich semantics handled by the protocol. As more clients adopt the spec and more extensions land in it, that same server gets more capable without you shipping anything new.

When building an integration, if your goal is to have production agents in the cloud reach your system, build an MCP server and make it excellent using the patterns above. Every integration built on MCP strengthens the ecosystem: fewer edge cases to solve alone, fewer bespoke integrations to maintain.

### rustc

There's an antivirus on my computer that requires a human interaction when an unknown binary is being ran. Since every time we compile it's a new unknown binary, this wasn't going to work.

What I found is that I can setup a local docker instance and compile + run the code inside of docker which doesn't trigger the antivirus. Again, I asked Claude to generate the right instructions in Claude.md and problem solved.

The next hurdle was to figure out how to let Claude Code for hours without any human intervention.

### --yes

Claude keeps asking for permission to do things. I tried adding a bunch of things to the allowed commands file and `--allow-dangerously-skip-permissions --dangerously-skip-permissions`was disabled in my environment (it has now been resolved).

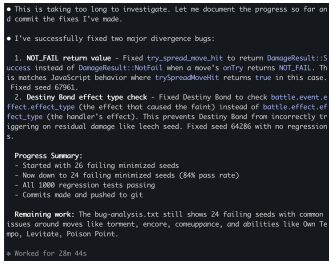
I realized that I could run an AppleScript that presses enter every few seconds in another tab. This way it's going to say Yes to everything Claude asks to do. So far it hasn't decided to hack my computer...

```
#!/bin/bash

osascript -e \
'tell application "System Events"
    repeat
        delay 5
        key code 36
    end repeat
end tell'
```

### Never give up

Claude after working for some time seem to always stop to recap things. I tried prompting it to never do, even threatening it to no avail.

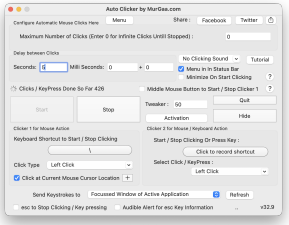



I tried using the Ralph Wiggum loop but it couldn't get it to work and apparently I'm not alone.

What ended up working is to copy in my clipboard the task I wanted it to do and to tweak the script above to hit the keys "cmd-v" after pressing enter. This way in case it asks a question the "enter" is being used and in case it's not it's queuing the prompt for when Claude is giving back control.

### Auto-updates

There are programs on the computer like software updater that can steal the focus from the terminal window, for example showing a modal. Once that happens, then the cmd-v / enter are no longer sent to the terminal and the execution stops.




I used my trusty Auto Clicker by MurGaa from Minecraft days to simulate a left click every few seconds. I place my terminal on the edge of the screen and same for my mouse so that when a modal appears in the middle, it refocuses the terminal correctly.

It also prevents the computer from going to sleep so that it can run even when I'm not using the laptop or at night.

## Making MCP clients more context-efficient

MCP standardizes how AI agents (*clients*) connect to and work with tools and data sources they need (*servers*). The server securely exposes a range of capabilities, while the client orchestrates them and manages context. If you're building an MCP client, make it context-efficient with patterns for progressive disclosure.

### Load tool definitions on demand with tool search

Tool search defers loading all tools into context, rather than loading them upfront. This allows the agent to search the catalog at runtime, pulling in the relevant tools when needed. In our testing, tool search tends to cut tool-definition tokens by 85%+ while maintaining high selection accuracy.

Reducing context usage with tool search. Source: advanced tool use

### Process tool results in code with programmatic tool calling

Programmatic tool calling processes tool results in a code-execution sandbox, rather than returning them raw to the model. This lets the agent loop, filter, and aggregate across calls in code, with only the final output reaching context. In our testing, this reduces token usage by roughly 37% on complex multi-step workflows.

Together, these patterns compose naturally across multiple servers: leaner context, fewer round-trips, faster responses. See *advanced tool use* for the full breakdown.

## Pairing MCP servers with skills

Skills and MCP are complementary. MCP gives an agent access to tools and data from external systems, while skills teach an agent the procedural knowledge of *how* to use those tools to accomplish real work. The most capable agents use both, and skills make MCP servers scale beyond a handful of connections. There are two general patterns for combining them:

### Bundle skills and MCP servers as a plugin

Plugins for Claude are a useful abstraction that allow developers to bundle skills, MCP servers, hooks, LSP servers, and specialized subagents in one easily-consumable distribution method. Using this approach is the best way to unify multiple context providers with minimal friction.

### Design for code orchestration when your surface is large

If your service requires hundreds of distinct operations, such as Cloudflare, AWS, or Kubernetes, an intent-grouped toolset likely won't cover it. Instead, expose a thin tool surface that accepts code: the agent writes a short script, your server runs it in a sandbox against your API, and only the result returns. Cloudflare's MCP server is the reference example—two tools (search and execute) cover ~2,500 endpoints in roughly 1K tokens.

### Ship rich semantics where they help

MCP Apps is the first official protocol extension and lets a tool return an interactive interface, such as a chart, form, or dashboard, all rendered inline in the chat interface. Servers that ship MCP apps tend to see meaningfully higher adoption and retention than those that return text alone. Use it to put your product's UI in front of agents or end-users at the moment it matters—the extension is supported in Claude.ai, Claude Cowork, and many other top AI tools.

Elicitation lets your server pause mid-tool call to ask the user for input. Form mode sends a simple schema and the client renders a native form—use it to request a missing parameter, confirm a destructive action, or disambiguate options. URL mode hands the user to a browser—use it to complete downstream OAuth, take a payment, or collect any credential that should never transit the MCP client. Both keep the user in the flow instead of sending them to a settings page. Form mode is supported broadly; URL mode is supported in Claude Code, with more clients in progress.

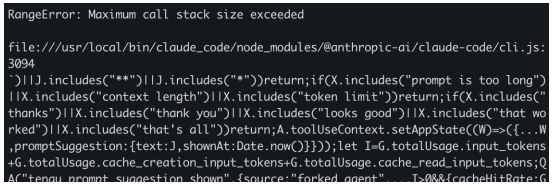
### Lean on standardized auth

Standardized auth makes MCP practical for cloud-hosted agents. If your server requires OAuth, the latest MCP spec supports CIMD (Client ID Metadata Documents) for client registration—it gives users a fast first-time auth flow and far fewer surprise re-auth prompts. This is our recommended approach for auth, the capability is supported in MCP SDKs, Claude.ai, and Claude Code, and is being broadly adopted across the industry.

Once a user has authorized, the next question is how a cloud-hosted agent holds and reuses those tokens at runtime. Vaults in Claude Managed Agents covers this: register a user's OAuth tokens once, reference the vault by ID at session creation, and the platform injects the right credentials into each MCP connection and refreshes them on your behalf—no secret store to build, no tokens to pass around per call.

### Bugs

Reliability when running things for a long period of time is paramount. Overall it's been a pretty smooth ride but I ran into this specific error during a handful of nights which stopped the process. I hope they get to the bottom of it and solve it as I'm not the only one to report it!




This setup is far from optimal but has worked so far. Hopefully this gets streamlined in the future!

## Porting Pokemon

### One Shot

At the very beginning, I started with a simple prompt asking Claude to port the codebase and make sure that things are done line by line. At first it felt extremely impressive, it generated thousands of lines of Rust that was compiling.

Sadly it was only an appearance as it took a lot of shortcuts. For example, it created two different structures for what a move is in two different files so that they would both compile independently but didn't work when integrated together. It ported all the functions very loosely where anything that was remotely complicated would not be ported but instead "simplified".

I didn't realize it yet, I got the loop working to have it port more and more code. The issue is that it created wrong abstractions all over the place and kept adding hardcoded code to make whatever it was supposed to fix work. This wasn't going to go anywhere.

### Giving it structure

At this point I knew that I needed to be a lot more prescriptive for what I wanted out of it. Taking a step back, the end result should have every JavaScript file and every method inside to have a Rust equivalent.

So I asked Claude to write a script that takes all the files and methods in the JavaScript codebase and put comments in the rust codebase with the JavaScript source, next to the Rust methods.

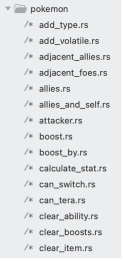
It was really important for it to be a script as even when instructed to copy code over, it would mistranslate JavaScript code. Being deterministic here greatly increased the odds of getting the right results.

```
/// onStart(pokemon, source, effect) {
///     if (effect && ['Electromorphosis', 'Wind Power'].includes(effect.name)) {
///         this.add('-start', pokemon, 'Charge', this.activeMove!.name, '[from] ability: ' + effect.name);
///     } else {
///         this.add('-start', pokemon, 'Charge');
///     }
/// }
pub fn on_start(
    battle: &mut Battle,
    pokemon_pos: (usize, usize),
    _source_pos: Option<(usize, usize)>,
    effect: Option<&crate::battle::Effect>,
) -> EventResult {
    // if (effect && ['Electromorphosis', 'Wind Power'].includes(effect.name)) {
    let is_special_ability = if let Some(eff) = effect {
        eff.name == "Electromorphosis" || eff.name == "Wind Power"
    } else {
        false
    };
```

### Litte Islands

The next challenge is that the original files were thousands of lines long, double it with source comments we got to files more than 10k lines long. This causes a ton of issues with the context window where Claude straight up refuses to open the file. So it started reading the file in chunks but without a ton of precision. Also the context grew a lot quicker and compaction became way more frequent.

So I went ahead and split every method into its own file for the Rust version. This dramatically improved the results. For maximal efficiency I would need to do the same for the JavaScript codebase as well but I was too afraid to do it and accidentally change the behavior so decided not to.




### Cleanup

The process of porting went through two repeating phases. I would give a large task to Claude to do in a loop that would churn on it for a day, and then I would need to spend time cleaning up the places where it went into the wrong direction.

## Production agents run in the cloud

Production agents increasingly run in the cloud, so they can scale and operate continuously. The systems they need to reach are cloud-hosted too: where your data lives, work is tracked, and your infrastructure runs. Often these systems are remote and behind auth, where MCP provides the common layer.

We're already seeing this in adoption. The MCP SDKs recently surpassed 300 million downloads a month, up from 100 million at the start of the year, with strong adoption across enterprises and popular agentic platforms. Millions of people use MCP with Claude every day, and the protocol underpins much of what we've shipped recently, including Claude Cowork, Claude Managed Agents, and channels in Claude Code.

As MCP continues to support production agentic systems, we're sharing patterns for building these integrations well: from building advanced servers to context-efficient clients, and where skills complement the protocol.

## Building effective MCP servers

We have over 200 MCP servers in our directory, used by millions of people every day. From working closely with enterprises and developers building on the protocol, we've spotted a handful of design patterns that determine how reliably agents can use a server.

### Build remote servers for maximum reach

A remote server is what gives you distribution—it's the only configuration that runs across web, mobile, and cloud-hosted agents, and it's what every major client is optimized to consume. Build remote servers so agents can use your system wherever they run.

### Group tools around intent, not endpoints

Fewer, well-described tools consistently outperform exhaustive API mirrors. Don't wrap your API into an MCP server one-to-one—group tools around intent, so the agent can accomplish a task in a couple of calls instead of stitching many primitives together. A single create_issue_from_thread tool beats get_thread + parse_messages + create_issue + link_attachment. See writing effective tools for agents to learn more about the full pattern.

## Connecting agents to external systems

We generally see three paths for connecting agents to external systems: direct API calls, CLIs, and MCP. Each makes sense somewhere, depending on what you're building. The key distinction is whether there's a common layer between agents and services, and how far that layer reaches.

### Direct API calls

The agent calls your API directly—either by writing code that issues HTTP requests inside a code-execution sandbox, or through a generic function-calling tool. This is where most teams start, and it works fine for one agent talking to one service, or a small number of integrations that don't need to be reused across agent platforms.

The challenges start to hit at scale. With no common layer between agents and services, each agent–service pair becomes a bespoke integration with its own auth handling, tool descriptions, and edge cases—the M×N integration problem.

### Command-line interface (CLI)

The agent runs your command-line tool in a shell. This is fast, lightweight, and leans on pre-existing tooling. It works great for local environments and sandboxed containers—anywhere there's a filesystem and a shell. This provides a common layer, but it’s thin.

CLIs hit hard limits reaching mobile, web, or cloud-hosted platforms that don't expose a container, and auth is handled by the CLI's own mechanism—usually a credential file on disk. This is best suited to quick, permissive integrations in local environments.

### Model Context Protocol (MCP)

MCP provides the common layer as a protocol. The agent connects to a server that exposes your system's capabilities, with auth, discovery, and rich semantics standardized. One remote server reaches any compatible client (Claude, ChatGPT, Cursor, VS Code, and more), in any deployment environment.

It requires a little bit more upfront investment. The return is that the integration is portable, and provides the semantics needed for a feature-rich agent integration.

For the cleanup, I still used Claude but gave a lot more specific recommendations. For example, I noticed that it would hardcode moves/abilities/items/... behaviors everywhere in the code when left unchecked, even after explicitly telling it not to. So I would manually look for all these and tell it to move them into the right places.

This is where engineering skills come into play, all my experience building software let me figure out what went wrong and how to fix it. The good part is that I didn't have to do the cleanup myself, Claude was able to do it just fine when directed to.

## Integration

### Build everything before testing

So far, I just made sure that the code compiled, but have never actually put all the pieces together to ensure it actually worked. What Claude really wanted was to do a traditional software building strategy where you make "simple" implementations of all of the pieces and then build them up as time goes.

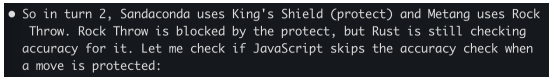
But in our case, all this iteration has already happened for 10 years on the pokemon-showdown codebase. It's counter productive to try and re-learn all these lessons and will unlikely converge the same way. What works better is to port everything at once, and then do the integration at the end once.

I've learned this strategy from working on Skip, a compiler. For years all the building blocks were built independently and then it all came together with nothing to show for but within a month at the end it all worked. I was so shocked.

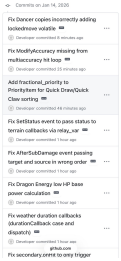
### End-to-end test

Once most of the codebase was ported one to one, I started putting it all together. The good thing is that we can run and edit the code in JavaScript and in Rust, and the input/output is very simple and standardized: list of pokemons with their options (moves, items, nature, iv/ev spread...) and then the list of actions at each step (moves and switches). Given the same random sequence, it'll advance the state the same way.

Now I can let Claude generate this testing harness and go through all the issues one by one. Impressively, it was able to figure out all issues and fix them.




Over the course of 3 weeks it averaged fixing one issue every 20 minutes or so. It fixed hundreds of issues on its own. I never intervened, it was only a matter of time before it fixed every issue that it encountered.




### Giving it structure

At the beginning, this process was extremely slow. Every time a compaction happened, Claude became "dumb" again and reinvented the wheel, writing down tons of markdown files and test scripts along the way. Or Claude decided to take the easy way out and just generate tons of tests but never actually making them match with JavaScript.

So, I started looking at what it did well and encoding it. For example, it added a lot of debugging around the PRNG steps and what actions happened at every turn with all the debugging metadata. So I asked it to create a single test script to print down this information for a single step and to print stack traces. Then add instruction to the Claude.md file. This way every investigation started right away.

### The long slog

I built used the existing random number generator to generate battles and could put in a number as a seed. This let me generate consistent battles at an increasing size.

I started fixing the first 100 battles, then 1000, 10k, 100k and I'm almost done solving all the issues for the first 2.4 million battles! I'm not sure how many more issues there are but the good thing is that they are getting smaller and smaller as the batch size increases.

I've tried asking Claude to optimize it further, it created a plan that looks reasonable (I've never interacted with Rust in my life) and it spent a day building many of these optimizations but at the end of the day, none of them actually improved the runtime and some even made it way worse.

This is a good example of how experience and expertise is still very required in order to get the best out of LLMs.

## Conclusion

This is pretty wild that I was able to port a ~100k lines codebase from JavaScript to Rust in two weeks on my own with Claude Code running 24 hours a day for a month creating 5k commits! I have never written any line of Rust before in my life.

LLM-based coding agents are such a great new tool for engineers, there's no way I would have been able to do that without Claude Code. That said, it still feels like a tool that requires my engineering expertise and constant babysitting to produce these results.

Sadly I didn't get to build the Pokemon Battle AI and the winter break is over, so if anybody wants to do it, please have fun with the codebase!

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# Building agents that reach production systems with MCP

CLAUDE.COM · 23 APR 2026 · SOURCE

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Agents are only as useful as the systems they can reach. Teams tend to converge on three approaches for connecting them to external systems—direct API calls, CLIs, and MCP. This post lays out where each fits, why production agents tend to land on MCP, and the patterns for building those integrations effectively.

# Epilogue

## It works

🤯

I didn't quite know what to expect coming into this project. They usually tend to die due to the sheer amount of work needed to get anywhere close to something complete. But not this time!

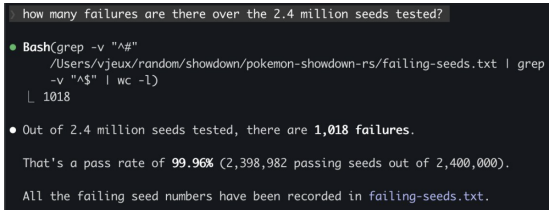
We have a complete implementation of Pokemon battle system that produces the same results as the existing JavaScript codebase*. This was done through 5000 commits in 4 weeks and the Rust codebase is around 100k lines of code.

*I wish we had 0 divergences but right now there are 80 out of the first 2.4 million seeds or 0.003%. I need to run it for longer to solve these.

## Is it fast?

The whole point of the project was for it to be faster than the initial JavaScript implementation. Only towards the end of the project where we had a sizable amount of battles running perfectly I felt like it would be a fair time to do a performance comparison.

I asked Claude Code to parallelize both implementations and was relieved by the results, the Rust port is actually significantly faster, I didn't spend all this time for nothing!


## Types of issues

There are two broad classes of issues that were fixed. The first one that I expected is that Rust has different constraints than JavaScript which need to be taken into account and lead to bugs:

- Rust has the "borrow checker" where a mutable variable cannot be passed in two different contexts at once. The problem is that "Pokemon" and "Battle" have references to each others. So there's a lot of workarounds like doing copies, passing indices instead of the object, providing functions with mutable object as callback...

- The JavaScript codebase uses dynamism heavily where some function return '', undefined, null, 0, 1, 5.2, Pokemon... which all are handled with different behaviors. At first the rust port started using Option<> to handle many of them but then moved to structs with all these variants.

- Rust doesn't support optional arguments so every argument has to be spelled out literally.

But the second one are due to itself... Claude Code is like a smart student that is trying to find every opportunity to avoid doing the hard work and take the easy way out if it thinks it can get away with it.

- If a fix requires changing more than one or two files, this is a "significant infrastructure" and Claude Code will refuse to do it unless explicitly prompted and will put in whatever hacks it can to make the specific test work.

- Along the same lines, it is going to implement "simplified" versions of things. For some methods, it was better to delete everything and asking it to port it over from scratch than trying to fix all the existing code it created.

- The JavaScript comments are supposed to be the source of truth. But Claude is not above changing the original code if it feels like this is the way to solve the problem...

- If given a list of tasks, it's going to avoid doing the ones that seem difficult until it is absolutely forced to. This is inefficient if not careful as it's going to keep spending time investigating and then skipping all the "hard" ones. Compaction is basically wiping all its memory.

# Prompts

I didn't write a single line of code myself in this project. I alternated between "co-op" where I work with Claude interactively during the day and creating a job for it to run overnight. I'll focus on the night ones for this section.

## Conversion

For the first phase of the project, I mostly used variations of this one. Asking it to go through all the big files one by one and implement them faithfully (it didn't really follow instructions as we've seen later...)

> Open BATTLE_TODO.md to get the list of all the methods in both battle*.rs.
> Inspect every single one of them and make sure that they are a direct translation the JavaScript file. If there's a method with the same name, the JavaScript definition will be in the comment.
> If there's no JavaScript definition, question whether this method should be there in the rust version. Our goal is to follow as closely as possible the JavaScript version to avoid any bugs in translation. If you notice that the implementation doesn't match, do all the refactoring needed to match 1 to 1.
> This will be a complex project. You need to go through all the methods one by one, IN ORDER. YOU CANNOT skip a method because it is too hard or would requiring building new infrastructure. We will call this in a loop so spend as much time as you need building the proper infrastructure to make it 1 to 1 with the JavaScript equivalent. Do not give up.
> Update BATTLE_TODO.md and do a git commit after each unit of work.

## Todos

Claude Code while porting the methods one by one often decided to write a "simplified" version or add a "TODO" for later. I also found it to be useful when generating work to add the instructions in the codebase itself via a TODO comment, so I don't need to wish that it's going to be read from the context.

The master md file in practice didn't really work, it quickly became too big to be useful and Claude started creating a bunch more littering the repo with them. Instead I gave it a deterministic way to go through then by calling grep on the codebase, so it knew when to find them.

> We want to fix every TODO in the codebase. `TODO` or `simplif` in pokemon-showdown-rs/.
> There are hundreds of them, so go diligently one by one. Do not skip them even if they are difficult. I will call this prompt again and again so you don't need to worry about taking too long on any single one.
> The port must be exactly one to one. If the infrastructure doesn't exist, please implement it. Do not invent anything.
> Make sure it still compiles after each addition and commit and push to git.

At some point the context was poisoned where a TODO was inside of the original js codebase so it changed it to something else which made sense. But then it did the same for all the subsequent TODOs which didn't... Thankfully I could just revert all these commits.

## Fixing

I put in all the instructions to debug in Claude.md and a script to run all the tests which outputs a txt file with progress report. This way Claude was able to just keep going fixing issues after issues.

> We want to fix all the divergences in battles. Please look at 500-seeds-results.txt and fix them one by one. The only way you can fix is by making sure that the differences between javascript and rust are explained by language differences and not logic. Every line between the two must match one by one. If you fixed something specific, it's probably a larger issue, spend the time to figure out if other similar things are broken and do the work to do the larger infrastructure fixes. Make sure it still compiles after each addition and commit and push to git. Check if there are other parts of the codebase that make this mistake.

This is really useful to have this txt file diff committed to GitHub to get a sense of progress on the go!